

Transforming clinical documentation with ambient artificial intelligence (AI) scribes: a narrative review of technology, impact, and implementation

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Background and Objective: Electronic health records (EHRs) have modernized care but increased documentation burden and clinician burnout. Ambient artificial intelligence (AI) scribes, combining automated speech recognition (ASR), natural language processing (NLP), and generative AI, aim to address this by capturing encounters and generating documentation. Related technologies, including virtual assistants and autonomous patient-facing systems, extend these capabilities beyond the clinician's physical presence. This narrative review synthesizes current evidence on the real-world performance, implementation, and impact of these AI tools.

Methods: A narrative literature search was conducted using PubMed, supplemented by a manual review of reference lists from key articles. The search covered studies published between January 2019 and June 2025. After screening and full-text review, 18 studies met inclusion criteria and were incorporated into this review.

Key Content and Findings: AI scribes consistently reduce documentation burden and cognitive load, improve workflow efficiency, save time, and enhance patient-clinician interaction by allowing greater clinician focus. However, studies also report frequent documentation omissions and occasional clinically significant hallucinations. Implementation remains a sociotechnical challenge involving workflow redesign, medico-legal considerations, and preservation of the patient-clinician relationship. In cardiology, where documentation requires precise, time-sensitive detail, AI-related errors may carry greater risk, underscoring the need for specialty-specific validation.

Conclusions: Ambient AI scribes show promise in reducing workload, improving efficiency, and decreasing burnout, but current systems still generate high omission rates and intermittent factual inaccuracies that may affect clinical decision-making. Evidence remains limited by small cohorts and methodological variability, warranting cautious interpretation. More rigorous, standardized evaluations are needed before routine clinical adoption.

Keywords: Artificial intelligence (AI); ambient artificial intelligence scribe (ambient AI scribe); accuracy; doctor-patient relationship; safety

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Submitted Aug 14, 2025. Accepted for publication Dec 30, 2025. Published online Jan 19, 2026.

doi: 10.21037/cdt-2025-454

View this article at: <https://dx.doi.org/10.21037/cdt-2025-454>

Introduction

Background

In the modern healthcare landscape, clinical documentation represents a paradox: it is both indispensable for quality care, communication, and legal compliance, yet it is also a primary source of profound professional clinicians' dissatisfaction and burnout (1). The widespread adoption of the electronic health record (EHR), intended to improve care through digital documentation, has not eased this burden. Instead, it has further bound clinicians to their screens. Time-and-motion studies show that for every hour of direct patient care, physicians spend about two hours on EHR and administrative tasks (1). This heavy clerical load has led to "pajama time", the after-hours work clinicians complete at home to finish documentation (2).

This persistent burden is not a mere inconvenience; it is a primary driver of the escalating crisis of clinician burnout. Defined as a syndrome of emotional exhaustion, depersonalization, and a low sense of personal accomplishment, it could lead to more medical errors, lower patient satisfaction, and higher physician turnover, threatening patient safety and workforce sustainability (3).

Healthcare systems used in-person and virtual medical scribes to reduce documentation burdens, but studies found that despite easing clerical work, the approach introduced significant challenges (4,5). These included substantial financial costs associated with hiring and training, high staff turnover, extensive training periods required to achieve proficiency, and frequent inconsistencies in note quality and style (4,5). Standard speech-recognition software, another alternative, often proved cumbersome but required line-by-line dictation and frequent, tedious correction, and it fundamentally failed to capture the unstructured, free-flowing, and conversational nature of a genuine clinical encounter (6).

Into this challenging environment enters the ambient artificial intelligence (AI) scribe, a technology designed to redefine clinical documentation. Ambient AI scribes passively listen to clinical encounters and automatically generate documentation, helping reduce workload, documentation time, and burnout while improving

patient-physician interaction and workflow efficiency (7). These advanced systems, such as Nuance's Dragon ambient eXperience (DAX), are powered by a sophisticated combination of automated speech recognition (ASR), natural language processing (NLP), and large language models (LLMs) (2,8). Operating "ambiently", they convert raw provider-patient conversation into structured notes, often in a standard format like SOAP (Subjective, Objective, Assessment, and Plan), with minimal clinician input (2,8).

Virtual assistants are another AI-driven tool designed to simulate human conversation and provide personalized health responses based on that. Capabilities range from simple menu or multiple choice-based assistants to more sophisticated conversational AI virtual assistants with NLP that recognize free speech or text (9). A patient-facing device is an agentic AI system designed to interact autonomously with patients. They can independently initiate actions based on contextual understanding and perform proactive workflow management and gather detailed histories and symptoms (10).

Rationale and knowledge gap

The promise of AI technology is immense and profound: to un-tether the clinician from the keyboard, restore face-to-face interaction, reduce the immense cognitive load of documentation, and reclaim precious personal time, thereby directly combating one of the primary drivers of burnout. However, as with any transformative technology in the high-stakes environment of healthcare, this promise is shadowed by critical questions of safety, accuracy, reliability, and equity. As healthcare organizations rapidly begin to pilot and adopt these technologies, a critical body of evidence is emerging from early-stage evaluations. Previous literature has examined the role of ambient AI in healthcare documentation; however, important gaps remain. Existing studies have not fully addressed several essential aspects, including detailed insights into the various AI systems, their use across different subspecialties, and a broader overview of all published work related to ambient AI scribes (11). In our paper, we also discuss the different agentic AI systems, further contributing to a more complete understanding of this evolving field.

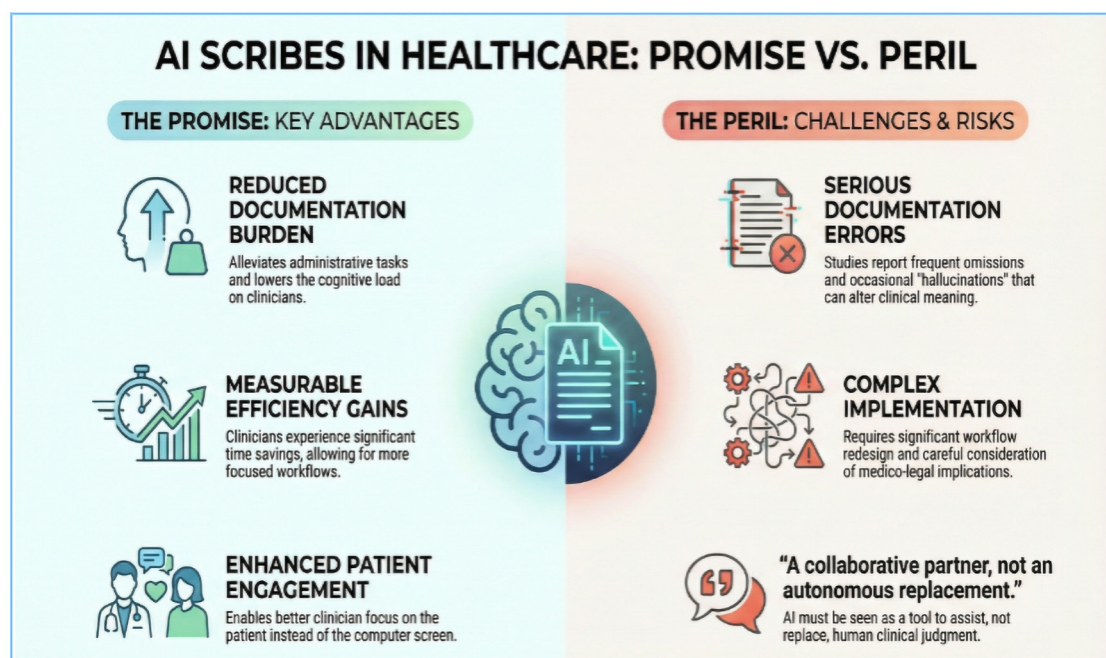


Figure 1 Graphical abstract: the impact of ambient AI scribes. AI, artificial intelligence.

Objectives

This narrative review seeks to explore the impact of AI scribes by examining their performance across five key domains: (I) the accuracy and safety of the documentation they produce; (II) their effect on clinician efficiency and workflow; (III) their influence on the clinician experience, particularly regarding burnout and well-being; (IV) their perceived impact on the patient-clinician relationship; and (V) the practical challenges and considerations for their real-world implementation. A visual summary is provided in the Graphical Abstract (*Figure 1*). We present this article in accordance with the Narrative Review reporting checklist (available at <https://cdt.amegroups.com/article/view/10.21037/cdt-2025-454/rc>).

Methods

We conducted a narrative literature search to identify relevant studies on the implementation and impact of ambient AI scribes in healthcare, querying PubMed and manually reviewing references from key articles to capture additional sources. The search spanned publications from January 2019 to June 2025, without restrictions on publication status, and was limited to English-language studies. Eligible designs included pilot trials, cohort studies, qualitative

research, simulation studies, and randomized controlled trials evaluating ambient AI scribe systems or comparable technologies in clinical practice. Exclusion criteria included review papers, conference abstracts, editorials, commentaries, preprints, and opinion pieces. Search terms combined keywords and Boolean operators, including "ambient scribe", "digital scribe", "artificial intelligence", "clinical documentation", and "speech recognition".

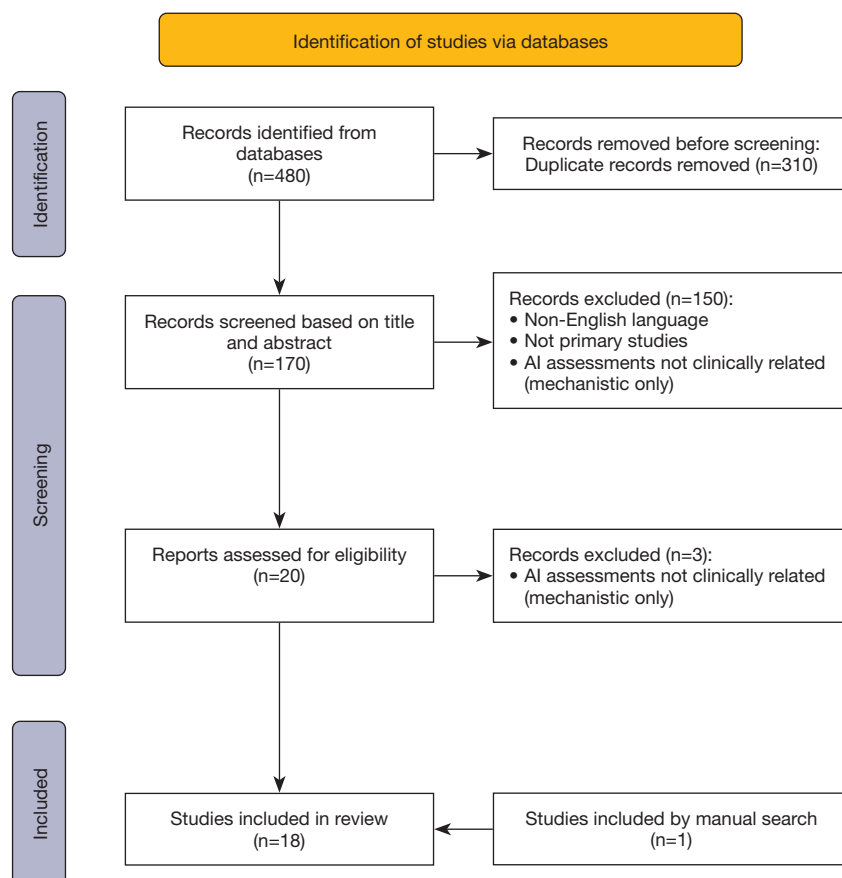
The literature search was last conducted on June 01, 2025, yielded a total of 480 records (310 duplicates). After screening 170 articles based on titles and abstracts by a single reviewer (with disagreements addressed through discussion with other authors if present), 20 articles were selected for full-text review. Following full-text evaluation, 17 articles met the inclusion criteria and were incorporated into this narrative review, and one additional article was identified through a review of the reference lists of the included studies. The complete search strategy and the PRISMA-style flow of study selection are presented in *Table 1* and *Figure 2*, respectively.

Main studies summarizing the current AI scribe knowledge

Recent studies spanning simulated and real-world

Table 1 Search strategy summary

Items	Specification
Date of search	06/01/2025
Database searched	PubMed
Search terms used	Search terms combined keywords and Boolean operators, including “ambient scribe”, “digital scribe”, “artificial intelligence”, “clinical documentation”, and “speech recognition”
Timeframe	January 2019 to June 2025
Inclusion and exclusion criteria	No restrictions on publication status. Exclusion criteria included review papers, conference abstracts, editorials, non-English language studies, commentaries, preprints, and opinion pieces. Eligible designs included pilot trials, cohort studies, qualitative research, simulation studies, and randomized controlled trials evaluating ambient AI scribe systems or comparable technologies in clinical practice
Selection process	The selection process was conducted by the first author. For articles where there was uncertainty regarding eligibility, the first author consulted with a co-author, who independently reviewed the article, and consensus was reached through discussion

**Figure 2** Flow diagram for searching of databases only.

outpatient settings across the U.S., U.K., and Australia have examined tools such as Nuance DAX, Abridge, ChatGPT-4, Nabla, and TORTUS (2,8,12-22). These interventions, implemented primarily in primary care and multispecialty contexts, were assessed through a mix of pre-post studies, qualitative methods, simulations, and limited randomized designs. Most tools demonstrated modest to significant reductions in documentation time (ranging from ~1 to 2.1 minutes per note), alongside improvements in perceived usability, clinician satisfaction, and reduced cognitive load or disengagement (2,8,12-22). Simulated environments often revealed high omission and hallucination rates in AI-generated notes, underscoring concerns about documentation accuracy and variability between vendors. While real-world integrations with systems like Epic and Cerner showed promise to enhancing workflow efficiency and user experience yet the evidence remains heterogeneous, with limitations such as small sample sizes, lack of randomization, and potential selection biases (2,8,12-22). Publication bias remains a key limitation, as unsuccessful or challenged AI scribe implementations are less likely to be reported, limiting insight into real-world feasibility. The unblinded design of most studies also introduces observer and performance bias, warranting cautious interpretation of benefits. Despite these issues, existing evidence suggests that well-designed AI tools have the potential to improve documentation and support clinician well-being. *Table 2* summarizes the main studies discussing the role of AI scribe in health care.

The quest for accuracy and safety

Clinical documentation tools must be accurate and safe, as errors in the medical record can cause diagnostic delays, treatment mistakes, and patient harm. Although AI scribes can capture more comprehensive encounters, their probabilistic design introduces new error risks, and evaluations show wide variability with potentially serious pitfalls.

This can be more important in cardiovascular field as the diagnosis and treatment often rely on subtle and time-sensitive clinical details, such as characterizing chest pain, documenting arrhythmia features, or recognizing early signs of heart failure. Documentation inaccuracies introduced by ambient AI scribes may carry heightened risks in this specialty. These highlight how even small documentation errors can have disproportionate consequences. Thus, future studies should focus on covering these aspects.

A sobering picture: high error rates and inconsistency

Two of the most detailed studies on AI scribe accuracy, conducted in simulated settings, paint a particularly sobering picture. One study by Kernberg *et al.* evaluated the performance of ChatGPT-4 in writing SOAP-format notes using a standardized template (“generate a clinical note in SOAP format for the following”) from 14 transcribed clinical encounters. The analysis revealed a startling average of 23.6 errors per clinical case (22). The most accurate section of the note was consistently the “Objective” section, which includes structured data like vital signs (median accuracy 86.9%), while the more narrative “History and Physical” and “Assessment and Plan” sections were significantly less accurate.

Similarly, Biro *et al.* carried out a meticulous assessment of two popular commercial ambient digital scribe (ADS) products using 44 simulated outpatient encounters. Their analysis identified a total of 127 errors, for an average of 2.9 errors per note (20). A crucial finding was that 70% of the notes generated contained at least one error. In this study there were 127 errors [mean 2.9, standard deviation (SD) 2.7 errors per draft note] in 31 of 44 (70%) draft notes. ADS product A resulted in 66 errors in 22 notes (mean 3.0, SD 2.7 per draft note) and product B resulted in 61 errors in 22 notes (mean 2.8, SD 2.7 per draft note). Product A had 55 omission errors (83%), 3 addition errors (4%), 4 wrong output errors (6%), and 4 irrelevant or misplaced text errors (6%). Product B had 33 omission errors (54%), 7 addition errors (11%), 6 wrong output errors (10%), and 15 irrelevant or misplaced text errors (25%). There was a statistically significant difference in error types between the two ADS products (Fisher exact test, $P=0.002$), underscoring that performance is not uniform across the market.

The anatomy of errors: omissions, fabrications, and factual mistakes

The most concerning and consistently observed issue across Kernberg and Biro *et al.* studies was the nature of the errors produced by the AI systems. Kernberg *et al.* classified these errors into three main categories. The most prevalent were omissions, accounting for 86.3% of all errors, where the AI frequently failed to include critical clinical information present in the original transcript, for example, omitting a patient’s loss of appetite in a hernia case or neglecting to mention echocardiogram results in a patient with congestive heart failure (22). The second most common error type was

Table 2 Overview of main studies

Study (year)	Setting/population	Intervention/AI tool	Study design	Sample size	Clinical context	Provider type	Comparator/control	Note/EHR integration	Outcomes measured	Time reduction	Burnout/satisfaction	Documentation error rate/type	Note quality	PX	Main findings
Kernberg <i>et al.</i> (2024)	Simulated ambulatory, U.S., 14 cases (3×)	ChatGPT-4 (SOAP notes)	Comparative, simulation	14 cases × 3 repetitions =42	Multispecialty	Simulated/Doctor of Medicine (MD)	Gold standard	No	Note accuracy, omission/addition, PDQI-9	N/A	N/A; simulated	86% omission, 23.6/case	PDQI-9 (average score 29.7)	N/A	High omissions, accuracy varies by case, ↓ accuracy with ↑ transcript length
Kocaballi <i>et al.</i> (2020)	GP, Australia	Prototype AI assistant	Qualitative, co-design	16 GPs	Primary care	MD (GP)	None	Video demonstration	Themes: AI role, workflow, trust	N/A	N/A	N/A	N/A	N/A	GPs prefer adaptive/human-in-loop, concern regarding medico-legal implications
Stults <i>et al.</i> (2025)	Outpatient, Sutter Health, U.S.	Abridge ambient scribe	Pre-post, quality QI, survey + EHR	100 clinicians, 57 paired	Multispecialty	MD, NP, APP	Baseline/pre	Integrated	Burnout, NASA-TLX, EHR time, after-hours	−0.9 min/pt	7% burnout↓, satisfaction↑	Not specified	Usability	↑	Changes in burnout are non-significant, but workflow is better, note time↓, NASA-TLX↓, primary care provider more positive
Liu <i>et al.</i> (2024)	Atrium Health/ Wake Forest (PCP, academic, U.S.)	DAX Copilot (Nuance/Microsoft)	Longitudinal cohort study	112 DAX, 103 control	Primary care, multispecialty	MD, DO and APP	Controls	Not available	EHR use metrics (EHR-Time, Note-Time, closure rates, note length), (wRVUs)	~7% documentation Hours↓	N/A	Not specified	Not specified	N/A	Objective gains for high DAX users only
Biro <i>et al.</i> (2025)	Simulated outpatient scripts, U.S.	2 commercial ADS (unnamed)	Simulated experiment	44 notes, 11 scripts	Multispecialty	Residents	None	No	Error frequency/type (omission, etc.)	N/A	N/A	70% of notes had errors, B 54% A 83% omission	N/A	N/A	High errors (omissions most) varies by vendor/product
Owens <i>et al.</i> (2024)	UM Health West, U.S.	Nuance DAX	Observational, survey/time	110 invited	Primary care	MD, NP, PA	Baseline	Integrated	OLBI, documentation time, PX	−1.8 min/note	Disengagement↓	Not quantified	Not specified	↑	Documentation time↓, less after-hours, disengagement improved, PX better (open label)
Albrecht <i>et al.</i> (2025)	U Kansas Med, 30 ambulatories. U.S.	Abridge ambient AI	Pre + post anonymous surveys	181 invited, ~93–99 pre/post	Multispecialty	MD, DO, APP	None	Some integration	Workflow ease, after-hours, completion, burnout	Not specified	67% less burnout	Not specified	Survey	↑	Workflow ease; 77% less after hours; 64% more satisfaction
Duggan <i>et al.</i> (2025)	Academic. outpatient, Philadelphia, 17 specialties, U.S.	DAX Copilot (Nuance, in EHR)	Single-arm pre-post QI, feedback	46 clinicians	Multispecialty	MD, NP, PA	Baseline	Epic	EHR time, note closure, after-hours	−2.1 min/note	Mental burden/engagement ↑	Not reported	Not specified	↑	Time-in-notes↓20%, after-hours↓30%, same-day closure↑
Shah <i>et al.</i> (2024)	Stanford Health, U.S. ambulatory	DAX Copilot (Nuance/Epic)	Prospective pre-post QI	48 MDs (38 paired)	PCP + specialists	MD only	None	Integrated/ smart sections	NASA-TLX, PFI (burnout), SUS, time	Median 20 min/half-day	PFI burnout↓, SUS↑	Not reported	SUS	↑	Task load↓, usability+, median 20 min saved/half-day
Haberle <i>et al.</i> (2024)	Intermountain Health, 12 specialties, U.S.	Nuance DAX ambient scribe	Peer-matched cohort	99 DAX, 76 controls	Multispecialty	MD, DO	Controls	Oracle Cerner	Document time/note, after-hours, safety, panel size	−0.76 min/note	Engagement↑, not full burnout	Not reported	Not reported	=	Documentation time per note↓, after-hours EHR slight↑, engagement↑, no PX change
Balloch <i>et al.</i> (2024)	U.K. simulated pediatric (mock clinics)	TORTUS ambient AI	Crossover, simulated	8 clinicians, 47 consults	Pediatrics (simulated)	MD, consultant	Simulated charting/EHR	Not clinical EHR	SAIL (document quality), time, NASA-TLX, PX	−193 s/consult	Task load↓, attention/focus↑	Minor/major hallucination (low%)	SAIL	Actor focus ↑	SAIL documentation quality↑, consults shorter-26%, task load↓, attention/focus↑
Kakaday <i>et al.</i> (2025)	Samaritan Health, operation room, U.S., PCP/urgent care	DAX Copilot (Nuance)	Randomized pilot	45 (25 DAX, 20 control)	PCP + urgent care + multispecialty	MD, NP, PA	Control	Epic	Document efficiency: time, note characters, % done by DAX	−1.4 min/visit	Not assessed	50% note characters by DAX (high users)	Not reported	N/A	Documentation time↓; high users: 50% characters by DAX
Misurac <i>et al.</i> (2025)	University of Iowa Health, U.S.; volunteers	Nabla ambient AI	Pre–post, 5wk	38 (35 completed)	Multispecialty	MDs only	Baseline	Not fully integrated	Stanford PFI (burnout), fulfillment	Not reported	Burnout↓69→43%, fulfillment not significant	Not detailed	Not reported	not significant	Burnout↓; fulfillment↑ (not significant)
Nguyen <i>et al.</i> (2023)	Moffitt Cancer, U.S.	Nuance DAX ambient	Pre/post survey + interviews	9 post-surveys, 8 interviews	Oncology/ oncologists	MD	None	Not specified	Mini-Z, sleep, usability, edit burden	Not measured	Burnout perception improved (qualitative)	Not specified	Not detailed	Some ↑	Perception: feasible, edit burden noted, not statistical burnout difference (scores)

ADS, ambient digital scribe; AI, artificial intelligence; APP, advanced practice provider; DAX, Dragon ambient eXperience; DO, Doctor of Osteopathic Medicine; EHR, electronic health record; GP, general practitioner; MD, Medical Doctor; N/A, not applicable; NASA-TLX, NASA Task Load Index; NP, nurse practitioner; OLBI, Oldenburg Burnout Inventory; PA, physician assistant; PCP, primary care physician; PDRQ-9, Patient-Doctor Relationship Questionnaire-9; PFI, Professional Fulfillment Index; PX, patient experience; QI, quality improvement; SAIL, Sheffield Assessment Instrument for Letters; SOAP, Subjective, Objective, Assessment, and Plan; SUS, System Usability Scale; U.K, United Kingdom; U.S, United States; wRVUs, work relative value unit.

additions or fabrications (10.5%), in which the AI generated content not present in the source, such as falsely stating that a patient's weight loss was intentional or that they were noncompliant with medications. The third and least frequent category was incorrect facts (3.2%), where the AI recorded existing clinical data inaccurately, such as reporting a normal heart rate when it was tachycardic or misstating the timing of hospital admission. Biro *et al.* confirmed a similar distribution of error types, with omissions again being most frequent (20). Supporting qualitative evidence from Bundy *et al.* revealed that 12 clinicians from Atrium Health -a multi-site academic health system- using DAX Copilot, reported issues such as the AI misgendering patients, generating inappropriate or unsolicited diagnoses, and confusing key clinical details, errors that ultimately required extensive manual correction, thereby diminishing the intended efficiency gains of the technology (12).

The insidious nature of omission errors

The predominance of omission errors is particularly concerning for patient safety. As Biro *et al.* pointed out, there was a fundamental cognitive difference in reviewing for different error types (20). Errors of commission (e.g., adding incorrect information) are often easier for clinicians to detect because they involve noticing information that is clearly wrong. Omissions, however, require recalling specific details from conversations hours or days earlier, making them harder to catch and more dangerous as they can silently propagate through the record.

The challenge of “hallucinations” and non-determinism

Beyond omissions, the phenomenon of “hallucinations”, fabricated details not present in the source pose serious risks, such as incorrectly linking a patient to the wrong medical test (20). These distortions can alter the clinical narrative and mislead providers.

Furthermore, the study by Kernberg *et al.* highlighted the critical issue of non-determinism. When the same clinical transcript was fed into ChatGPT-4 three separate times, the model produced three different notes with varying errors. An alarmingly low 52.9% of the data elements were consistently and correctly reported across all three versions (22). This lack of reproducibility prevents clinicians from anticipating where errors will appear, undermining reliable oversight across different sections of the note.

A contrasting view: evidence of high-quality documentation

However, the narrative on accuracy is not entirely negative and contains important contradictions. In a simulated clinical encounter, Balloch *et al.* evaluated a GPT-4 powered ambient AI system created by TORTUS. The simulation consisted of a structured, controlled outpatient consultation involving real clinicians and professional medical actors following standardized patient scripts, with clinicians blinded to the scripted scenarios (18). Using the validated Sheffield Assessment Instrument for Letters (SAIL), researchers found that AI-generated notes scored higher in quality than those produced through standard EHR workflows, suggesting that, when functioning properly, AI scribes can create more complete and better-structured documentation than time-pressured clinicians.

Likewise, in a cohort study evaluating 99 physician interactions using Nuance DAX, Haberle *et al.* reported that no patient safety events associated with DAX were identified in their safety event tracking system, suggesting a strong safety profile in that specific real-world implementation (2). A more neutral view was presented by Duggan *et al.*, who demonstrated that clinicians had varying perceptions of the accuracy and completeness of notes from DAX Copilot in a study including 46 providers, suggesting a mixed but not overtly negative experience (19).

The clinician's role as final guarantor

These findings, taken together, illustrate a clear but complex pattern. While some tools and studies demonstrate promise in improving documentation quality, others reveal frequent and dangerous errors. The ultimate conclusion is that while these tools are powerful, they are not yet infallible. The responsibility for the final accuracy and safety of the medical record remains non-negotiable with the clinician, making the process of diligent, meticulous review an essential and unavoidable part of the AI-assisted workflow.

Reclaiming time: the impact on clinician efficiency and workflow

The primary value proposition of AI scribes is their potential to save clinicians time, thereby improving efficiency and alleviating the after-hours documentation burden (8,19). The evidence largely supports this claim, though the magnitude and nature of the time savings are more nuanced and complex than often portrayed, revealing

several key paradoxes and differences in impact (8,19).

The core success: quantifiable reductions in documentation time

Multiple studies utilizing objective EHR data have consistently shown a statistically significant reduction in clinician documentation time following the implementation of AI-based tools, although the extent of this reduction varies. Duggan *et al.* conducted a quality improvement (QI) study with 46 clinicians using Nuance DAX Copilot and reported a 20.4% decrease in time spent on notes per visit, from 10.3 to 8.2 minutes ($P<0.001$) (19). Similarly, Owens *et al.* observed a 28.8% reduction (approximately 1.8 minutes per note), among high-frequency users ($\geq 60\%$ of encounters) in a community teaching health system (8). Stults *et al.* (2025), evaluating the Abridge AI tool with 57 clinicians, found a significant drop in documentation time per encounter from 6.2 to 5.3 minutes ($P<0.001$). In a peer-matched controlled trial, Haberle *et al.* reported that Nuance DAX reduced documentation time per visit from 5.3 to 4.54 minutes ($P<0.001$) (2). Balloch *et al.* in a simulated pediatric consultation using TORTUS AI, found a 193-second (26.3%) reduction in average consultation duration ($P=0.03$) (18). Lastly, a pilot study by Cao *et al.* (2024) conducted in dermatology clinics with 12 clinicians evaluating DAX, demonstrated a decrease in total daily documentation time from 54.6 to 42.2 minutes (23). Collectively, these findings indicate that ambient AI documentation tools are associated with meaningful, statistically significant improvements in documentation efficiency across various settings.

Subjective perceptions of efficiency

These quantitative findings are further reinforced by qualitative data and clinician self-reports, which consistently highlight perceived improvements in documentation efficiency. In a survey by Albrecht *et al.* evaluating the Abridge tool, clinicians were nearly five times more likely to agree that they could complete their notes before the next patient encounter [odds ratio (OR) 4.95; 95% confidence interval (CI): 2.87–8.69; $P<0.001$], with 77% reporting a reduced documentation burden (14). Similarly, Shah *et al.* found that 65% of respondents reported enhanced efficiency in documentation tasks, estimating a median time saving of 20 minutes per half-day clinic session (24). Supporting these findings, Nguyen *et al.*

reported a significant increase in clinicians' perception of having adequate time for documentation following the implementation of Nuance DAX, with relevant survey scores improving from 2.1 to 3.6 within one month ($P=0.005$) (15). These self-reported experiences align closely with objective metrics, underscoring the positive impact of AI-based documentation tools on clinician workflow.

The “pajama time” paradox: a deep dive into after-hours work

AI scribe tools reduce active daytime documentation, but their effect on after-hours EHR use, or “pajama time”, is more complex and reveals a key tension in current clinical workflows. Several studies report notable improvements: Duggan *et al.* reported a 30.0% decrease in after-hours charting, from 50.6 to 35.4 minutes daily ($P=0.02$). While Owens *et al.* reported an 11.8% decline, equating to 4 fewer minutes per day (95% CI: 1–7.2) (8,19). Similarly, Albrecht *et al.* found that 73% of clinicians using the Abridge reported reduced documentation outside scheduled clinical hours (14). However, these benefits are not universal. A study by Haberle *et al.* including 99 providers across multiple outpatient clinics in Utah found a significant increase in after-hours EHR usage among AI scribe users, with a 4.69% rise ($P<0.05$), contrasting with a decline in the control group (2). The authors suggest that this may be due to the delayed availability of finalized notes in systems involving a human-in-the-loop for quality review, requiring clinicians to return to the EHR later to finalize documentation. Additionally, Stults *et al.* reported no statistically significant change in after-hours documentation ($P=0.14$), suggesting a neutral effect (13). This “pajama time paradox” highlights that while AI scribes may lessen documentation time burden in clinics, delays in note availability can offset benefits, underscoring the need for near-instantaneous documentation.

“Note bloat”: the double-edged sword of comprehensive capture

Another paradox with AI-based documentation tools is their tendency to increase clinical note length even as they reduce creation time. Duggan *et al.* reported a 20.6% rise in total weekly note length from 202,637.5 to 244,427.1 characters ($P<0.001$) (19). Similarly, Owens *et al.* found that although documentation became more efficient, the

average note length increased by 542 characters, even as the manual input from providers decreased by 33% (8). Stults *et al.* also observed a statistically significant increase in both total documentation and progress note length ($P=0.01$ and $P<0.001$, respectively) (13). Further complicating this trend, Kernberg *et al.* noted substantial variability in the length of AI-generated notes, even across replicates of the same clinical case (22). This phenomenon, often referred to as “note bloat” which can enhance billing and preserve detailed clinical context but may obscure essential information, hinder rapid understanding of patient status and care plans, and increase clinicians’ cognitive burden overall.

The “dose-response” relationship: heterogeneity of impact

The most nuanced understanding of AI scribe efficiency comes from studies showing that benefits vary across users. Liu *et al.*, in a longitudinal study, involving 215 outpatient clinicians at Atrium Health, reported no statistically significant overall efficiency gains among all DAX users (25). However, high-frequency users (>60% of encounters) showed modest reductions in documentation time. Similarly, the STREAMLINE pilot by Kakaday *et al.* reported that clinicians using DAX CoPilot in $\geq 70\%$ of problem-focused visits had a 1.4-minute reduction per visit and a 35% decrease in note length, though not statistically significant ($P=0.38$), likely due to limited sample size (17). Together, these findings suggest a “dose-response” relationship in which the advantages of AI scribe technology become evident primarily with consistent, intensive integration into clinical workflows, indicating that infrequent or casual use is unlikely to produce meaningful time savings.

The clinician experience: alleviating burnout and enhancing well-being

Beyond the technical metrics of time and accuracy, a key measure of an AI scribe’s value lies in its impact on medical providers. The evidence strongly suggests that, despite performance limitations, AI scribes are having a positive effect in this domain by reducing cognitive burden and improving professional well-being.

Measuring the unseen: reductions in cognitive load

Clinical documentation imposes substantial cognitive burden as clinicians balance patient interaction, decision-making, and note-taking. AI scribes help alleviate this

load by “offloading” documentation. Stults *et al.*, using the NASA Task Load Index (NASA-TLX), reported significant reductions in mental demand (12.2 to 6.3), feelings of a rushed pace (13.2 to 6.4), and overall effort needed to complete notes (12.5 to 7.4) (13). Similarly, Balloch *et al.* reported improvements across five of six NASA-TLX workload dimensions in their simulation study (18). Complementing these quantitative findings, qualitative interviews conducted by Bundy *et al.* reveal the human experience behind the data: physicians expressed profound relief through “cognitive offloading”, no longer burdened by the anxiety of retaining critical clinical information until they had time to document it. One clinician described the alleviation of the “burned-out feeling” and the helpless anxiety associated with recalling clinical details after a demanding day (12). Together, these insights underscore AI scribes’ potential to mitigate the cognitive challenges inherent in clinical documentation.

A direct assault on burnout: evidence from validated instruments

The reduction in cognitive load associated with AI scribe use appears to support measurable improvements in clinician well-being and burnout, though results differ across studies. Shah *et al.* reported that among 38 physicians using DAX Copilot, work-related exhaustion decreased significantly, with scores on the Stanford Professional Fulfillment Index’s Work Exhaustion subscale (PFI-WE) dropping by 1.94 points ($P<0.001$) (24). Similarly, Misurac *et al.* found a significant decline in burnout with Nabla, with overall scores decreasing from 4.16 to 3.16 ($P=0.005$) and improvement in “interpersonal disengagement” (16). In contrast, pilot studies by Stults *et al.* and Albrecht *et al.* involving the Abridge AI tool with 57 clinicians showed a reduction in burnout rates from 42.1% to 35.1%, though this change did not reach statistical significance ($P=0.12$) (13,14). Owens *et al.*, in an observational study of 110 primary care providers, found that those with high DAX usage (>60% of visits) exhibited significantly lower disengagement scores ($P=0.03$), despite no significant differences in exhaustion or overall burnout (8). Contradicting these positive trends, Nguyen *et al.* reported a slight, non-significant increase in burnout scores (Mini Z scale) from 3.6 to 3.9 among cancer care providers using DAX Copilot ($P=0.08$), though qualitative feedback remained favorable (15). Collectively, these findings suggest that while AI scribes may help reduce burnout and improve

well-being, effects are heterogeneous and may depend on clinical context and user engagement.

The tangible feeling of improved job satisfaction

Even when composite burnout scores did not significantly change, reduced administrative burden consistently improved clinician job satisfaction. Albrecht *et al.* reported that 64% of clinicians agreed the AI tool improved their work satisfaction (14). Similarly, Stults *et al.* found that 71.9% of clinicians experienced increased job satisfaction, with primary care providers benefiting the most (85.8%) compared to medical (36.4%) and surgical (50.0%) subspecialists ($P<0.001$) (13). Galloway *et al.* observed a significant decline in the negative impact of documentation on clinician well-being, decreasing from 71% to 38.7% ($P=0.01$) among 31 surveyed clinicians. Collectively, these findings show AI scribes reduce documentation burdens and help restore fulfillment in clinical practice.

Perspectives on the patient-clinician encounter

Although AI scribe research often centers on clinicians, its effect on patients and the patient-clinician relationship is crucial. Introducing an ambient listening device raises concerns about privacy and trust, yet evidence indicates that thoughtful implementation can enhance engagement and support a more personable clinical encounter, fostering empathy.

Removing the “third party”: restoring the dyad

AI scribes may strengthen the patient-provider relationship by removing the computer as a physical and cognitive barrier. Stults *et al.* showed that Abridge use increased clinicians' full attentiveness from 57.9% to 93.0% ($P<0.001$), with clinicians feeling less distracted and more engaged (13). Supporting these findings, Duggan *et al.* similarly reported improved engagement, with disengagement scores decreasing from 5.41 to 2.05 and charting distraction decreasing from 5.67 to 2.27 ($P<0.001$) (19). Qualitative findings from Bundy *et al.* emphasized enhanced presence and eye contact (12). Similarly, Nguyen *et al.* reported that the ambient AI scribe “removes the computer as the third party”, improving patient-physician connection (15). Together, these findings underscore AI scribes' significant role in enhancing the quality and intimacy of clinical encounters.

The patient's perspective: a favorable, if understudied, view

The impact of AI scribes on the patient experience remains a crucial yet underexplored area, with the existing evidence generally positive but underscoring the importance of transparency. Owens *et al.* conducted a notable two-phase study that highlighted this dynamic: during the open-label phase, when patients were aware that AI technology was being used, over 75% reported that providers appeared more focused, typed less, and made the encounter feel more personal, all statistically significant improvements (26). Conversely, in the masked phase, where patients were unaware of the technology, no significant difference in Patient-Doctor Relationship Questionnaire-9 (PDRQ-9) scores was observed (PDRQ-9 median of 45 in both groups; $P=0.31$), illustrating the importance of patient perception. Supporting these findings, Balloch *et al.* reported that 87% of patients felt fully attended to when AI tools were utilized, compared to 75% with the EHR alone (18), while Galloway *et al.* found that 91.4% of patients agreed providers spent less time on the computer (27). Haberle *et al.* (2024) documented a remarkably low patient opt-out rate of just 0.014% (5 patients), indicating strong patient acceptance when the technology is transparently presented, though they observed no significant change in patient satisfaction as measured by Likelihood to Recommend scores ($P=0.49$) (2). Earlier, Kocaballi *et al.* raised concerns that patients might devalue physician recommendations if perceived as computer-generated, highlighting the delicate trust essential to the therapeutic relationship (21). While current research has not confirmed this fear, it emphasizes that AI scribes must be positioned clearly as tools designed to augment rather than replace human connection and clinical judgment to maintain patient trust and satisfaction.

Specialty-specific variations in ambient AI impact

The impact of ambient AI documentation systems varies markedly across clinical specialties. Studies show significant differences in clinician satisfaction, documentation time, and perceived workflow efficiency (13,14,25), likely reflecting specialty-specific documentation demands.

Primary care reported the highest satisfaction, with 85% (33/38) noting improved work experience, compared with only 36.4% (4/11) in medical subspecialties such as oncology, cardiology, and dermatology, and 50% (4/8) in surgical subspecialties ($P<0.001$) (13). Although ambient AI greatly improved workflow ease overall (OR 6.91;

Table 3 Impact of ambient AI across clinical specialties

Metric category	Specialty subgroup finding	AI platform	Estimate (95% CI) or P value	First author
Efficiency: documentation time	Primary care experienced a significant decrease in mean time in notes per appointment	Abridge	Mean decreased from 6.3 to 5.2 minutes, $P<0.001$	Stults <i>et al.</i>
Efficiency: documentation time	Family medicine showed an exploratory decrease in documentation hours (Note-Time) compared to the control group	DAX Copilot	MR 0.91 (0.85 to 0.98)	Liu <i>et al.</i>
Efficiency: documentation time	Internal medicine showed no statistically notable difference in documentation hours (Note-Time) compared to the control group	DAX Copilot	MR 1.03 (0.93 to 1.15)	Liu <i>et al.</i>
Efficiency: documentation time	Medical subspecialties spent significantly longer on notes per appointment compared to primary care	Abridge	Mean increase of 3.75 minutes (2.09 to 5.41 minutes), $P<0.001$	Stults <i>et al.</i>
Efficiency: documentation time	Surgical subspecialties spent significantly less time on notes per appointment than primary care	Abridge	Mean reduction of 2.45 minutes (−4.10 to −0.81 minutes), $P=0.004$	Stults <i>et al.</i>
Efficiency: workflow metrics	Pediatrics showed a significant increase in same day note closure rate	DAX Copilot	Mean difference 2.92% (1.07 to 4.76)	Liu <i>et al.</i>
Efficiency: workflow metrics	Family medicine showed a small, significant increase in completed appointment rate	DAX Copilot	Mean difference 1.02% (0.20 to 1.83)	Liu <i>et al.</i>
Documentation detail: note length	Medical subspecialties experienced the largest increase in document length post-AI implementation compared to primary care	Abridge	Mean increase of 2,323.72 characters (994.70 to 3,652.73 characters), $P=0.001$	Stults <i>et al.</i>
Well-being: work satisfaction	Primary care participants reported significantly higher increased work satisfaction post-AI compared to subspecialties	Abridge	OR (medical/surgical vs. primary care): 0.02 (<0.01 to 0.16), $P<0.001$	Stults <i>et al.</i>
Well-being: subjective impact	Clinicians' perception of improved documentation workflow ease and reduced burnout risk did not differ significantly by specialty type	Abridge	Analysis showed no significant effect of specialty type on survey responses	Albrecht <i>et al.</i>

AI, artificial intelligence; CI; confidence interval; DAX, Dragon ambient eXperience; MR, means ratio; OR, odds ratio.

95% CI: 3.90–12.56; $P<0.001$) (14), specialists were far less likely than primary care clinicians to report increased work satisfaction (OR 0.02; 95% CI: <0.01–0.16) (13). Subspecialists frequently attributed this gap to poor specialty-specific customization, especially in physical examination sections that lacked neurologic or cardiologic detail and required manual revision.

Efficiency gains also varied. Primary care achieved the largest reduction in documentation time (6.3 to 5.2 minutes per appointment; $P<0.001$). In contrast, medical subspecialists spent 3.75 minutes more per appointment than primary care ($P<0.001$). Family medicine showed modest improvements [means ratio (MR) 0.91; 95% CI: 0.85–0.98], whereas internal medicine (MR 1.03; 95% CI: 0.93–1.15) and pediatrics (MR 0.98; 95% CI: 0.90–1.08) showed none. Low-volume DAX users had small reductions

(MR 0.91; 95% CI: 0.83–0.99) (25).

Ambient AI also increased note length across specialties, with medical subspecialties producing the longest notes and adding 2,323.72 characters compared with primary care ($P=0.001$) (13). This additional documentation burden again reflected limited customization and the need for specialists to manually adjust autogenerated examination content.

Overall, ambient AI yields greater satisfaction and efficiency in primary care, while subspecialties experience reduced benefit due to poor alignment with their documentation requirements. No studies have evaluated ambient AI scribes specifically in cardiology, highlighting a clear research gap and the need for specialty-focused assessments of performance, workflow integration, and clinician experience in cardiovascular practice. A summary is provided in *Table 3*.

From pilot to practice: real-world challenges and future directions

Translating a promising technology from a controlled pilot study to widespread, effective practice is a formidable challenge. Implementation hurdles are deeply sociotechnical, rooted in workflow, user behavior, and organizational context. Current performance likely reflects early technological maturity in this domain. Continued advancements in workflow adaptability, seamless health record integration, and model accuracy are anticipated as these tools evolve. As clinicians gain familiarity and confidence, both performance and satisfaction should improve.

Sociotechnical hurdles to widespread adoption

Clinician experience with AI scribes shows that adoption is not uniform. Bundy *et al.* reported wide variability in preferences, with some clinicians favoring AI scribes for simple, structured tasks and others for complex, narrative-heavy encounters (12). In oncology, Nguyen *et al.* found that AI scribes supported history-taking but performed poorly for patient education and required extensive editing, emphasizing the need for specialty-specific customization (15). Early implementations also created workflow friction due to multi-step recording and manual EHR transfer. While some users experienced improvement, verbose or disorganized outputs increased editing workload, reinforcing the necessity of seamless EHR integration to minimize clicks and context switching (12,14).

Bundy *et al.* also identified a “productivity paradox”, where clinicians feared that time saved by AI scribes would be repurposed to increase patient volume rather than reduce workload, potentially worsening burnout (12). Medico-legal concerns further complicated adoption. Kocaballi *et al.* reported clinician anxiety over permanent, word-for-word AI-generated records that could be used in litigation, especially if statements were taken out of context (21). This underscores a broader tension between comprehensive documentation and protecting clinical judgment and professional autonomy.

The health system return on investment (ROI) remains uncertain. Although 58.1% of clinicians in Galloway *et al.* felt productivity improved (27) and 48% in Albrecht *et al.* believed they could see additional patients (14), objective data do not confirm these perceptions. Liu *et al.* found no significant change in work relative value units (wRVUs) or revenue per visit (25), and Haberle *et al.* similarly noted

stable panel sizes with only minimal, non-significant wRVU increases. Thus, while AI scribes may meaningfully improve clinician wellness and reduce burnout-related turnover, direct financial gains remain unclear. High implementation costs further complicate the business case, prompting exploration of secondary benefits such as enhanced documentation quality for coding and reimbursement, though financial impact is still uncertain (28).

Critical gaps and future research imperatives

Several critical gaps and future directions define the evolving landscape of AI scribe technology. First, there is an urgent need for standardized metrics across studies. Currently, research employs a heterogeneous mix of burnout inventories: such as the Stanford Professional Fulfillment Index (PFI), Oldenburg Burnout Inventory (OLBI), and Mini Z alongside various task load scales like NASA-TLX, usability assessments, and custom EHR analytics, complicating direct comparisons and meta-analyses.

Health equity remains underexplored, with no rigorous evaluations of how AI scribes perform for diverse patients, including those with accents, dialects, or interpreters (29). This raises substantial concerns that models trained predominantly on majority-group data may underperform for marginalized populations, potentially exacerbating existing healthcare disparities. Specifically, within cardiovascular field, it is essential to conduct studies that validate these tools across diverse patient populations and clinical settings. Such efforts will help ensure that AI-driven documentation systems are both generalizable and equitable, supporting accurate, efficient, and patient-centered care across the full spectrum of cardiovascular practice.

Looking ahead, the maturation of AI scribes is envisioned to progress beyond current capabilities. Seth *et al.* characterize existing tools as operating primarily at Stage 1 or 2 focused on automating clinical and administrative documentation (30). The future, however, lies in Stages 3 and 4, where AI evolves into a reactive and eventually proactive clinical decision support system. Kapadia *et al.* describe this advanced model as the “AI-augmented clinician” or “co-pilot”, where the AI not only documents but also actively supports care by alerting clinicians to missed screening questions, suggesting differential diagnoses based on dialogue, and assisting with ordering and referral workflows (31). Agentic AI in patient-facing devices offers a proactive, structured alternative to ambient

listening by engaging patients directly and asynchronously. These agents can gather histories and symptoms before visits, giving clinicians more comprehensive context and ensuring no critical questions are missed, improving documentation and billing justification. By offloading routine data collection, agentic AI allows clinicians to focus on reasoning and human connection, strengthening patient interactions. Rather than simple automation, this approach aims to elevate care quality and efficiency. This shift could further transform clinical practice by enhancing workflow efficiency and supporting more informed decision-making

Strengths and limitations

In our study, we sought to comprehensively review all available literature and contextualize previously reported findings to better define the potential benefits and limitations of available software, thereby informing how this technology could be optimally implemented in clinical practice. Additionally, our review performed a detailed evaluation of the various AI systems and their use across different subspecialties.

However, as a narrative review dependent on primary sources, our analysis is inherently limited by the scope and quality of those studies, which include the potential for observer bias and the generally small sample sizes reported in several studies. Additionally, the scarcity of literature addressing specific subspecialty implications restricts our ability to draw robust conclusions for these areas. Furthermore, the use of different AI models across studies each trained on distinct datasets makes direct, head-to-head comparisons challenging.

Conclusions

The rise of ambient AI scribes has laid important groundwork in reducing documentation burdens and clinician burnout. However, important challenges persist, including inconsistent performance, omission errors, note bloat, and variability related to differences in training datasets, study designs, and small sample sizes. These issues underscore the continued need for human oversight and cautious clinical integration. Current systems also remain largely passive, relying on clinician-directed encounters and limiting their broader transformative potential. Progress will require combining agentic AI with ambient technologies to enable proactive, structured data capture and real-time contextual reasoning.

Acknowledgments

None.

Footnote

Reporting Checklist: The authors have completed the Narrative Review reporting checklist. Available at <https://cdt.amegroups.com/article/view/10.21037/cdt-2025-454/rc>

Peer Review File: Available at <https://cdt.amegroups.com/article/view/10.21037/cdt-2025-454/prf>

Funding: None.

Conflicts of Interest: All authors have completed the ICMJE uniform disclosure form (available at <https://cdt.amegroups.com/article/view/10.21037/cdt-2025-454/coif>). C.A. serves as an unpaid editorial board member of *Cardiovascular Diagnosis and Therapy* from April 2025 to March 2027. C.A. also serves as Section Editor with Stipend for *Digital Medical, Circulation: Cardiovascular Imaging*. The other authors have no conflicts of interest to declare.

Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Cite this article as: Razaghi M, Hafez A, Farina JM, Scalia IG, Pereyra M, Abdelfattah FE, Sheashaa H, Awad K, Lester SJ, Ayoub C, Arsanjani R. Transforming clinical documentation with ambient artificial intelligence (AI) scribes: a narrative review of technology, impact, and implementation. *Cardiovasc Diagn Ther* 2026;16(1):11. doi: 10.21037/cdt-2025-454